Unit: Unit 0: Foundations of Algebra
Topic: Inequalities

Name: ____________
Date: __________

Foundation (Jalapeno)

Q1. Solve the inequality $2x > 6$
(A) $x > 3$
(B) $x < 3$
(C) $x = 3$
(D) $x \geq 3$
(E) $x \leq 3$

Q2. Graph the solution to $x < 2$ on a number line
(A) All numbers to the left of 2
(B) All numbers to the right of 2
(C) All numbers equal to 2
(D) All numbers greater than or equal to 2
(E) All numbers less than or equal to 2

Q3. Write the solution to $x > -4$ in interval notation
(A) $(-\infty, -4)$
(B) $(-4, \infty)$
(C) $[-4, \infty)$
(D) $(-\infty, -4]$
(E) $[-4, -4]$

Q4. Solve the inequality $x + 2 > 5$
(A) $x > 3$
(B) $x < 3$
(C) $x = 3$
(D) $x \geq 3$
(E) $x \leq 3$

Q5. Graph the solution to $x \geq 0$ on a number line
(A) All numbers to the left of 0
(B) All numbers to the right of 0
(C) All numbers equal to 0
(D) All numbers greater than or equal to 0
(E) All numbers less than or equal to 0

Q6. Write the solution to $x < 1$ in interval notation
(A) $(-\infty, 1)$
(B) $(1, \infty)$
(C) $[1, \infty)$
(D) $(-\infty, 1]$
(E) $[1, 1]$

Q7. Solve the inequality $3x < 12$
(A) $x < 4$
(B) $x > 4$
(C) $x = 4$
(D) $x \geq 4$
(E) $x \leq 4$

Q8. Graph the solution to $x \leq -2$ on a number line
(A) All numbers to the left of -2
(B) All numbers to the right of -2
(C) All numbers equal to -2
(D) All numbers greater than or equal to -2
(E) All numbers less than or equal to -2

Open Ended Questions (FRQ)

Q9. Solve the inequality $2x + 5 > 11$ and write the solution in interval notation

Q10. Graph the solution to $x > -3$ and $x < 2$ on a number line and write the solution in interval notation

Chef's Tips

- Check for understanding of inequality concepts
- Emphasize the importance of graphing and interval notation
- Provide feedback on problem-solving strategies